

J. Baeden Vester

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EDUCATION

North Carolina State University

Master of Computer Science

Raleigh, NC

Aug. 2025 - Present

- **Coursework:** Software Engineering, Robotics, Operating Systems Fundamentals, Cybersecurity

University of North Carolina at Chapel Hill

Bachelor of Science in Computer Science, Minor in Chinese

Chapel Hill, NC

Aug. 2022 - May 2025

- **Coursework:** Algorithms, Data Structures, Databases, Models of Languages and Computation, Computer Organization, Computer Vision, Numerical Analysis

EXPERIENCE

Product Developer

Sep. 2023 – Present

CircuitSays

Raleigh, NC

- Developed a **React-based** companion web app for physical electronics kits, improving user engagement and product usability, leading to **10 new client schools** beginning the grant process to acquire our electronics kits as well as **over \$20,000** in standalone grants.
- Assisted in the design and development of physical kits to teach circuit fundamentals, enhancing learning outcomes through hands-on educational tools.
- Implemented a testing suite with **80% code coverage** to ensure a reliable and professional-grade application for our educational clients.
- Collaborated with potential clients to gather feedback, using insights from the app to demonstrate product value and improve product-market fit.

PROJECTS

Multi-Drone Control | *Python, Isaac Sim, Isaac Lab, PX4, MAVLink, QGroundControl, Pegasus*

- Built a modular simulation framework in **NVIDIA Isaac Sim** for simultaneous programmatic control of multiple drones using both high-level flight commands and low-level motor control.
- Implemented dual control architectures supporting **MAVLink commands** via PX4 and QGroundControl for teleoperation alongside a **custom PID controller** for direct motor-level control.
- Integrated **Isaac Lab** reinforcement learning policies to enable heterogeneous multi-agent simulations combining drones with other robot types in a shared environment.
- Designed an extensible template architecture with modular documentation covering PX4 controller creation, direct motor control, and RL environment training workflows.

CalculateTripCost - Driving Expense Tracker | *React, FastAPI, PostgreSQL, Tailwind CSS, Chart.js*

- Engineered a full-stack web application with **React frontend** and **FastAPI backend** to track and analyze driving expenses with real-time cost calculations.
- Integrated with the **U.S. Energy Information Administration API** to fetch current gas prices by state and implement automatic price caching.
- Designed and implemented a **PostgreSQL database schema** with optimized queries for statistical analysis and monthly reporting.
- Developed interactive data visualizations using **Chart.js** to display driving statistics, cost breakdowns, and monthly trends.
- Built a responsive UI with **Tailwind CSS** featuring modern design patterns and seamless mobile/desktop compatibility.

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, Bash, Java, JavaScript, HTML/CSS, SQLite

Frameworks/Libraries: React, Node.js, Vite, FastAPI, NumPy, PyTorch, Matplotlib, OpenCV, Langchain

Developer Tools: Linux, Neovim, Git, Make, Docker, VS Code, ChatGPT, Claude, Ollama